

Additional Practices on “One to One” association relationship

Practice: Library Management System

A library needs a system to manage members and their issued library cards. Each library member must have a unique library card that includes attributes such as card number, issue date, and expiration date.

The system should:

1. Verify the library card is valid and not expired.
2. Track the total number of books a member has borrowed and enforce a borrowing limit of 5 books at a time.
3. Calculate the number of days left until the card expires and send a renewal reminder.

Practice: School Management System

A school needs a system to manage students and their unique ID cards. Each student must have an ID card that includes attributes such as card number, issue date, and grade level.

The system should:

1. Verify the ID card's validity and ensure it corresponds to the current academic year.
2. Track the student's attendance percentage and flag them if their attendance falls below 75%.
3. Calculate the remaining duration of validity for the ID card and notify students for renewal.

Practice: Hotel Booking System

A hotel requires a system to manage room bookings and their associated booking receipts. Each room booking must have a one-to-one relationship with a receipt containing details like receipt ID, booking date, and total amount paid.

The system should:

1. Ensure the receipt corresponds to a valid and confirmed booking.
2. Track the check-in and check-out dates associated with the receipt.
3. Calculate the total stay duration (in days) using the check-in and check-out dates and validate that the payment matches the duration of stay.

Practice: E-commerce Order Management System

An e-commerce platform requires a system to manage orders and their associated invoices. Each order must have a unique invoice that includes attributes such as invoice number, order date, and total amount.

The system should:

1. Validate that the invoice matches the correct order and calculate taxes (e.g., 15%) based on the total amount.
2. Track the order's payment status (paid, pending, or failed) using the invoice details.
3. Generate a summary of the order, including the invoice date and time, to share with the customer.

Practice: University Course Management System

A university needs a system to manage professors and their assigned office rooms. Each professor must be associated with a unique office room that includes attributes like room number, building name, and capacity.

The system should:

1. Validate that the room is assigned to only one professor at a time.
2. Track the professor's name, department, and contact details linked to the room.
3. Calculate the total number of office hours the professor spends weekly and ensure it adheres to the university's policy.

Practice: Pet Ownership System

A pet adoption center requires a system to manage pets and their adoption certificates. Each adopted pet must have a one-to-one association with a certificate containing details

like certificate number, adoption date, and adopter's name.

The system should:

1. Verify the adoption certificate's authenticity and ensure it corresponds to a registered pet.
2. Track the pet's species, breed, age, and vaccination status using the certificate details.
3. Calculate the pet's time with the adopter (in months) based on the adoption date and flag overdue vaccinations.